

original $\sum_{i=1}^n y_i w^T x_i$

↓

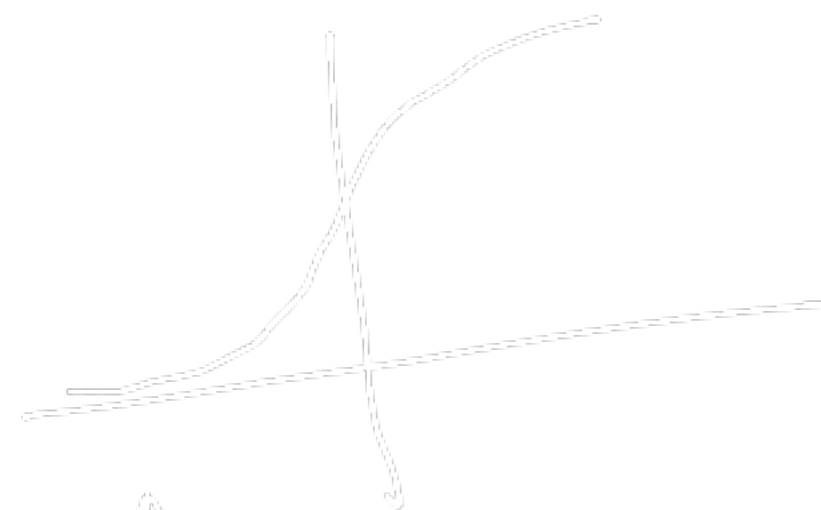
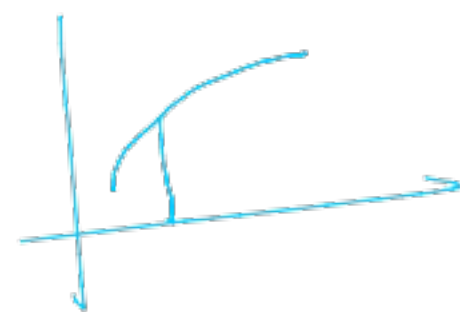
sigmoid fn

original $\sum_{i=1}^n \sigma(y_i w^T x_i)$

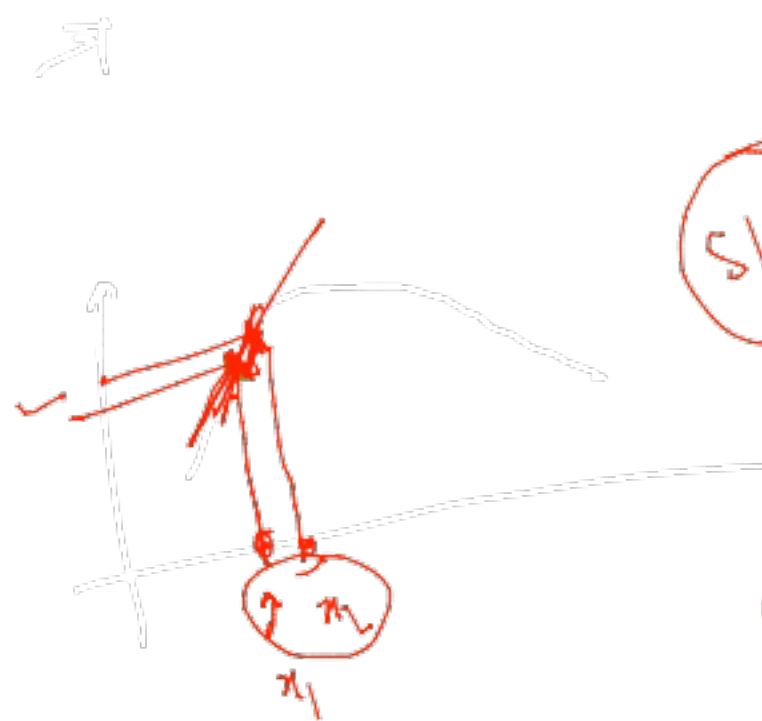
↑

monotonical fn
 $\Downarrow$
 g(u)

slope



$$y = \frac{1}{1 + e^{-x}}$$



slope = $\frac{\Delta y}{\Delta x}$

gradients
 $\downarrow$
derivative

$$y = \sin u$$

$$\frac{dy}{du} = \cos u$$

derivative $\frac{dy}{du} = \cos u$

$$\Rightarrow w^* \argmax \sum_{i=1}^n \frac{1}{1 + \exp(-y_i w^T x_i)}$$

$$g(x) = \log x$$

$$w^* \argmax \sum_{i=1}^n \log(\sigma(y_i w^T x_i))$$

$$\log\left(\frac{m}{n}\right) = \log m - \log n$$

$$\uparrow$$

$$\log m \quad \log n$$

$$\argmax_{w^*} \sum_{i=1}^n \log \left(\frac{1}{1 + \exp(-y_i w^T x_i)} \right)$$

const
 $\uparrow$

$$\argmax_{w^*} \sum_{i=1}^n \underbrace{\log 1}_{\uparrow 0} - \log(1 + \exp(-y_i w^T x_i))$$

$$\arg \max_w \sum_{i=1}^n \log (1 + \exp(-y_i \omega^T x_i))$$

↑
= f(w)

⇓

Note

$$\arg \max_{w^*} \underbrace{f(x)}_{\uparrow} = \arg \min_{w^*} f(x)$$

High

$$\left. \begin{array}{c} \omega \rightarrow \infty \\ \uparrow \\ \omega \rightarrow -\infty \end{array} \right\}$$

~~min~~

$$\arg \min_w \sum_{i=1}^n \log (1 + \exp(-y_i \omega^T x_i))$$

⇓
overfitting / underfitting
↳

→ optimisation
problems

$$\underline{y_i \omega^T x_i > 0}$$

L1 and L2 Regularization

$$\arg \min \sum_{i=1}^d \log(1 + \exp(-y_i \omega^T x_i))$$

let $z_i = y_i \omega^T x_i$

$$\arg \min \sum_{i=1}^d \log(1 + \exp(-z_i))$$

$$\sum_{i=1}^d \log(1 + \exp(-z_i))$$

ie. $\log(1 + \exp(-z_i)) > 1$ at $z_i = ?$

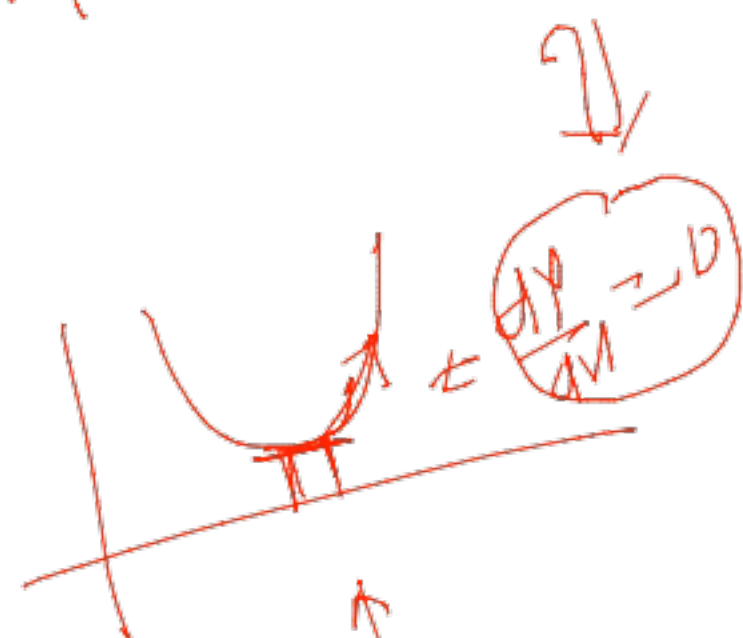
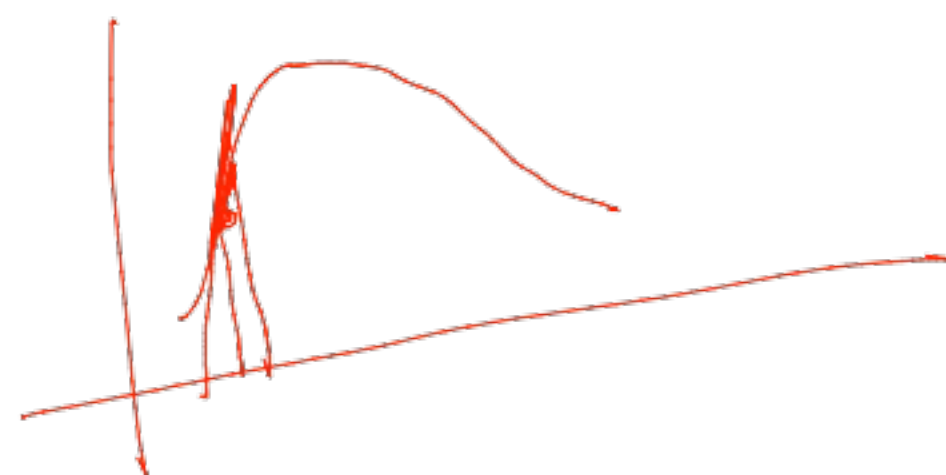
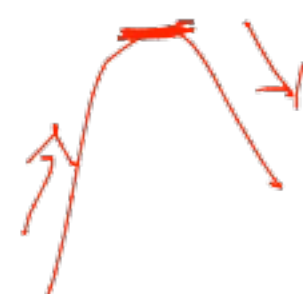
$$\text{plot}(\exp(-z_i))$$

$(+ve)$

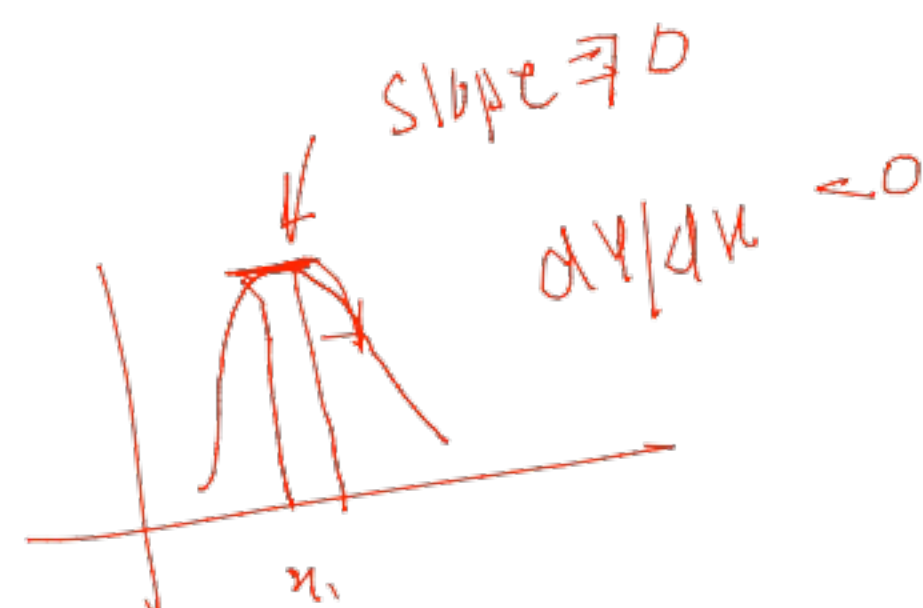
ω^T

if $\frac{dy}{dx} = 0$

slope



critical point or
turning point



$$w^* = \underset{w}{\operatorname{argmin}} \sum_{i=1}^d \log(1 + \exp(-z_i)) + \lambda \cdot \underline{\underline{w^T \cdot w}}$$

Hyper.

$\lambda \cdot \|w\|$
 (L1)
 ↑
 penalty

$\lambda \cdot \|w\|_2$
 (L2)
 ↑
regularization

sklearn → $\epsilon = 1$
 ↑
 (L2) → overfitting

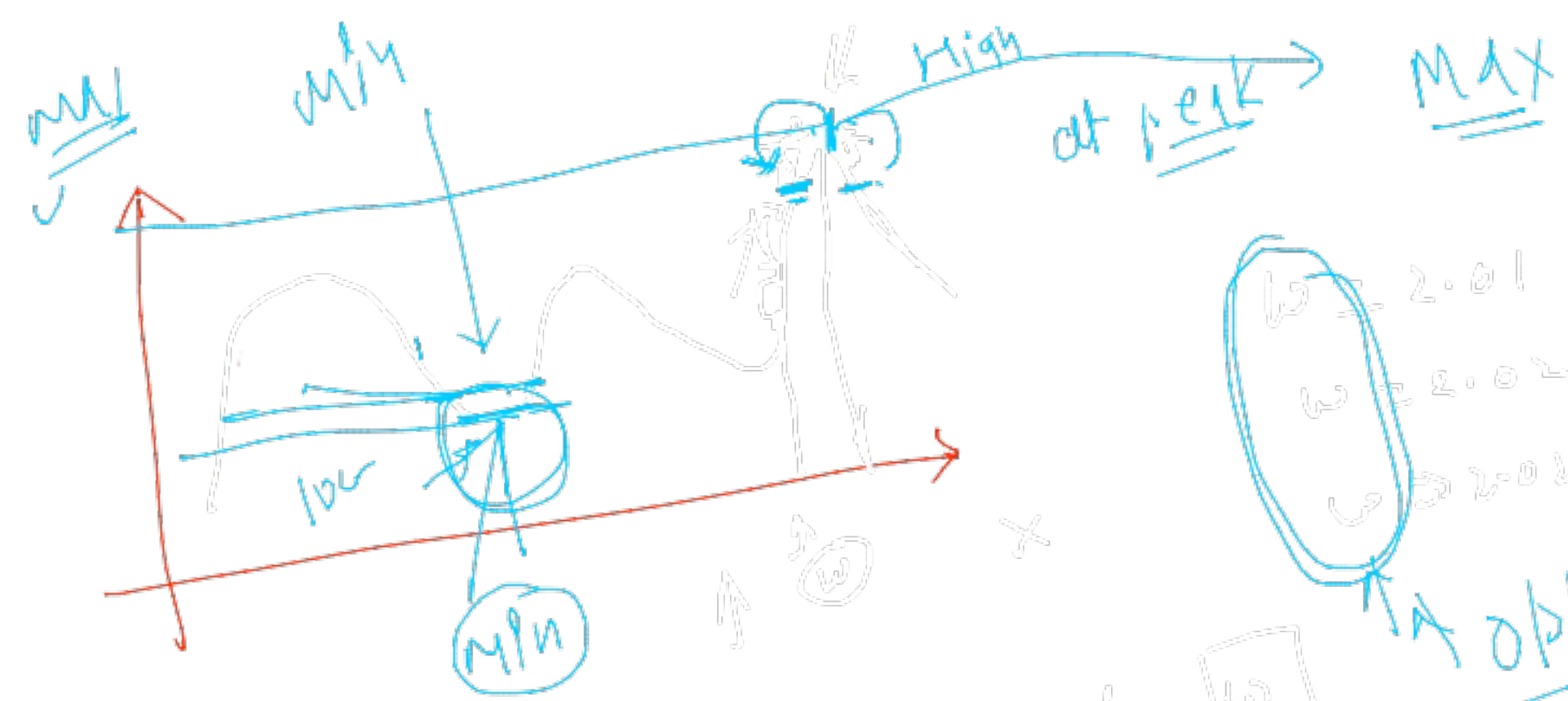
↑ error

↓ loss

minimizing

$y = f(x)$

learning rate = 0.01



$w = 2.01 \rightarrow 0.124$
 $w = 2.02 \rightarrow 0.1245$
 $w = 2.03 \rightarrow 0.1255$

optimization

diffⁿ value of w

$w = 1.01$
 $w = 1$
 $w =$

$\sigma(y - w^T(x))$
 feed

